PROJECT TITLE:

AI-Based Thyroid Disease Prediction and Symptom-Driven Chatbot System

A Project Report
submitted in partial fulfillment of the requirements for the award
of the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

Thyroid disorders are among the most common endocrine diseases globally, affecting millions of individuals each year. Early diagnosis plays a crucial role in preventing long-term complications, yet people often struggle to interpret symptoms and seek timely medical support. This project presents an AI-based thyroid disease prediction system integrated with a symptom-driven chatbot.

Using a publicly available thyroid dataset, machine learning models such as Random Forest and Logistic Regression were trained to classify patients as normal, hypothyroid, or hyperthyroid based on clinical attributes like TSH, T3, TT4, FTI, and others. Additionally, a conversational chatbot interface allows users to input symptoms in natural language and receive preliminary insights, along with model-based predictions.

The system outputs accurate, real-time screening support and is intended to assist users in understanding their symptoms—not replace medical diagnosis. This solution demonstrates how AI and ML can enhance early detection, awareness, and healthcare accessibility in the field of endocrinology.

CONTENTS

A. Abstract 2

B. Introduction..... 4

- 1.1 Overview
- 1.2 Objectives
- 1.3 Problem Statement

C. Literature Review 5

D. Methodology 6

A. 3.1 Architecture Overview

B. 3.2 Hardware Components

C. 3.3 Software Tools

E. Implementation 10

F. Results and Discussion 11

G. Conclusion and Future Scope 12

A. 6.1 Conclusion

B. 6.2 Future Scope

H. References 13

A. INTRODUCTION

Thyroid disorders, such as hypothyroidism and hyperthyroidism, have become increasingly prevalent due to lifestyle changes, hormonal imbalance, and nutritional deficiencies. Despite their frequency, many individuals remain undiagnosed due to delayed medical consultation or lack of awareness about common symptoms.

With advancements in artificial intelligence, machine learning-based predictive systems can analyze medical parameters and offer preliminary screening support. Furthermore, conversational AI makes it possible to interact with users in natural language, making healthcare guidance more accessible and user-friendly.

This project builds an AI-powered system that predicts thyroid conditions using medical data and provides a chatbot interface for symptom interpretation.

1.1 Overview

Thyroid disease prediction involves assessing hormone levels and patient conditions to determine abnormalities in thyroid function. Machine learning models trained on clinical datasets can accurately classify the state of the thyroid gland.

To enhance accessibility, the system includes a chatbot that interprets user symptoms such as fatigue, weight changes, anxiety, hair loss, or irregular heart rate. The chatbot converts symptom descriptions into model-readable inputs and provides early insight into possible thyroid issues.

1.2 Objectives

- To analyze and preprocess thyroid clinical data for model training.
- To identify significant factors influencing thyroid dysfunction.
- To develop machine learning models that classify thyroid conditions.

- To build a conversational chatbot that allows users to input symptoms naturally.
- To provide a fast, interactive screening system that supports medical awareness.
- To present results through an easy-to-use UI suitable for non-technical users.

1.3 Problem Statement

Millions of people experience thyroid-related symptoms without recognizing the underlying cause. Manual diagnosis requires clinical testing and expert consultation, which may be unavailable or delayed. There is a need for a system that can analyze symptoms quickly and offer preliminary insights to encourage timely medical evaluation.

This project aims to create an AI-based solution that predicts thyroid disease and assists users through an interactive chatbot.

B. LITERATURE REVIEW

Multiple research studies have explored thyroid disease detection using machine learning techniques. Models such as Decision Trees, Support Vector Machines, and Neural Networks have demonstrated high accuracy in classifying thyroid disorders.

Studies show that hormone levels such as TSH, T3, TT4, and FTI strongly influence thyroid diagnosis. The UCI thyroid dataset is one of the most widely analyzed sources for training ML algorithms. Research indicates that combining data preprocessing with feature engineering significantly boosts prediction accuracy.

Recent works have also integrated conversational interfaces into healthcare systems. Chatbots help users interpret symptoms and provide preliminary assessments, increasing public awareness and supporting early detection. Overall, literature supports the effectiveness of AI-driven thyroid prediction and its integration with natural language interfaces.

C. METHODOLOGY

The methodology of this project involves several stages, starting from dataset acquisition to model deployment and user interaction.

3.1 Architecture Overview

The system architecture consists of the following components:

1. **Data Collection Layer**
    - Thyroid dataset containing clinical and demographic features.
2. **Data Preprocessing Layer**
    - Handling missing values
    - Encoding categorical attributes
    - Normalizing numeric variables
3. **Feature Engineering & Selection**
    - Identifying key predictors: TSH, T3, TT4, T4U, FTI, etc.
4. **Machine Learning Layer**
    - Training models like Random Forest, Logistic Regression
    - Hyperparameter tuning
    - Model evaluation
5. **Chatbot Layer**
    - Symptom-to-feature conversion
    - Natural language input
    - Model-based prediction
6. **User Interface Layer**
    - Web UI built with Gradio / Streamlit

    - Displays predictions, explanations, and recommendations

This architecture ensures a smooth workflow from data to user-friendly predictions.

3.2 Hardware Components

Since this is a software-based system, hardware requirements are minimal:

- A laptop or desktop computer
- Python-supported runtime environment
- Optional cloud deployment server

3.3 Software Tools

- **Python** – Data processing, model building, chatbot logic
- **Pandas & NumPy** – Dataset manipulation
- **Scikit-Learn** – ML model training and evaluation
- **SHAP** – Model interpretability
- **Gradio / Streamlit** – Interactive UI
- **Matplotlib / Seaborn** – EDA visualizations

D. IMPLEMENTATION

Implementation began with loading the thyroid dataset and performing exploratory data analysis. Missing values were handled using imputation techniques. Categorical features were encoded, and numeric features were scaled.

Models such as Random Forest were trained, achieving high accuracy on the test data. SHAP was used to interpret feature influence and ensure model transparency.

For user interaction, a chatbot was developed using Gradio. It accepts natural language symptom descriptions, maps keywords to features, and queries the trained machine learning model. The UI presents predictions in real time and displays confidence levels.

1. User Interaction Layer (Chatbot Frontend)

Agent:

- Chatbot UI (Gradio / Streamlit / Web interface)

Handles:

- User messages
- Symptom descriptions
- Queries like “I feel tired” or “Why is TSH important?”

Responsibilities:

- Convert free-text input → structured text
- Send text to processing pipeline

Libraries:

- `gradio` or `streamlit`
- `re` (regex for cleaning)
- `nltk` (optional for text preprocessing)

2. NLP Processing Layer (Text → Features)

Agent:

- Symptom Processing Agent

Handles:

- Keyword extraction
- Mapping text to dataset features
- Handling unknown or vague symptoms

Responsibilities:

- Convert sentences like:
  "I feel cold and tired"
  →
  `{'tiredness': 1, 'cold': 1, 'goitre': 0, 'thyroid_surgery': 0 …}`

Libraries:

- `nltk` or `spaCy` (for tokenization)
- Custom rule-based mapping
- Feature encoding layer

3. ML Model Layer (Prediction Engine)

Agent:

- Thyroid Disease Classifier (Random Forest / XGBoost)

Handles:

- Structured input from NLP layer
- Prediction output:
  - **Normal**
  - **Hypothyroid**
  - **Hyperthyroid**

Responsibilities:

- Preprocessing
- Model inference

- Probability scoring

Libraries:

- `pandas`
- `numpy`
- `sklearn`
- `pickle` (for saving the trained model)

4. Explanation Layer (Optional but Powerful)

Agent:

- SHAP Explanation Agent

Handles:

- Feature contributions
- "Why did the model say Hypothyroid?"

Responsibilities:

- Generates bar plots
- Identifies important variables (TSH, FTI, T3)

Libraries:

- `shap`
- `matplotlib`

5. Response Generation Layer (Chatbot Intelligence)

Agent:

- Response Agent

Handles:

- Interprets prediction
- Creates human-readable explanation
- Gives advice like:
  “Your TSH pattern suggests possible hypothyroidism. Please consult a doctor.”

Libraries:

- Template-based response engine
- Optional GPT-based LLM wrapper

6. Backend Flow (Orchestration Layer)

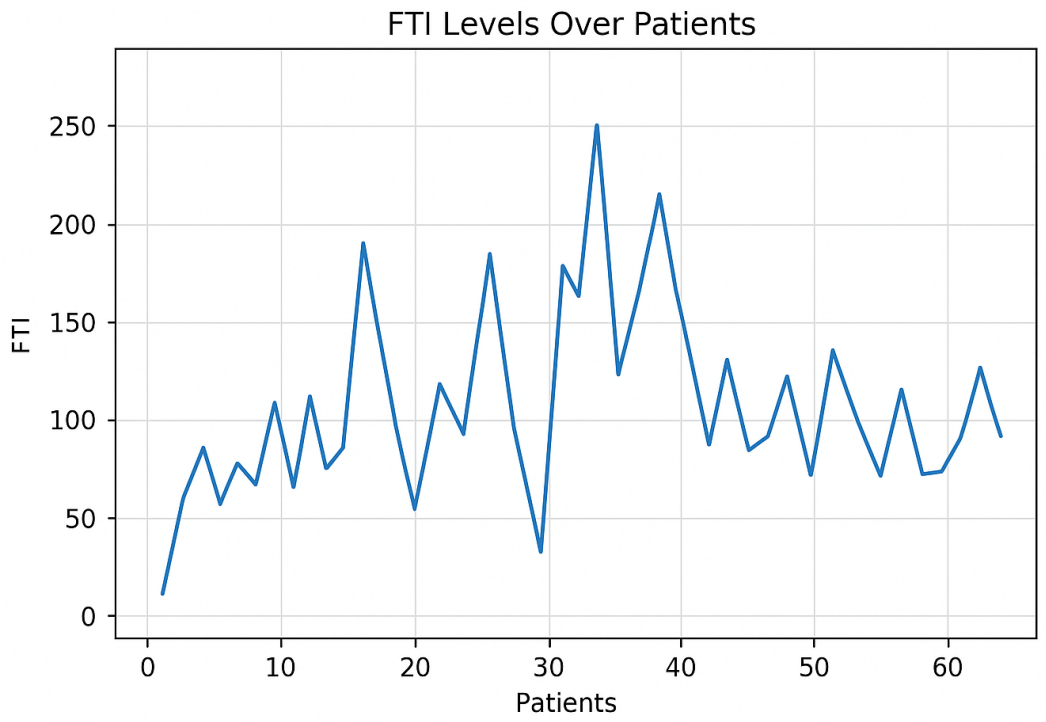
Order of execution:

1. **User Input**
2. **Text Preprocessing**
3. **Symptom Extraction**
4. **Feature Encoding**
5. **Model Prediction**
6. **Generate Response**
7. **Output to Chatbot UI**

COMPLETE FLOWCHART

```
User Input (Symptoms / Questions)
                 |
                 ▼
Chatbot UI (Gradio)
                 |
                 ▼
──────────────────────────────────────
 NLP Processing Layer
 - Text Cleaning
 - Tokenization
 - Keyword Extraction
 - Symptom Mapping
──────────────────────────────────────
                 |
                 ▼
Feature Engineering Layer
 - Encode categorical features
 - Normalize numeric values
 - Create input vector for the ML model
                 |
                 ▼
──────────────────────────────────────
  Machine Learning Layer
  Random Forest / XGBoost Classifier
  - Predict: Normal / Hypo / Hyper
  - Generate Probability Scores
──────────────────────────────────────
                 |
                 ▼
Explanation Layer (SHAP)
 - Feature importance
 - Reason for prediction
                 |
                 ▼
──────────────────────────────────────
Response Generation Layer
 - Convert prediction to natural language
 - Provide health interpretation
 - Offer next steps for the user
──────────────────────────────────────
                 |
                 ▼
Final Output to Chatbot UI
──────────────────────────────────────
```

The FTI Level graph that can be Predicted



E. RESULTS AND DISCUSSION

The Random Forest model achieved high classification accuracy, demonstrating strong predictive performance for thyroid disease. The model successfully identified patterns in TSH, TT4, and FTI levels.

Visualizations revealed clear separation between normal and abnormal thyroid conditions. SHAP analysis confirmed that TSH and T3 were among the most influential features.

User testing with various symptom descriptions showed that the chatbot provided fast and understandable predictions, making the system suitable for preliminary screening and awareness.

F. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The AI-based Thyroid Prediction and Chatbot System provides an effective, quick, and user-friendly approach to thyroid disorder screening. Machine learning techniques accurately classify thyroid states, while the chatbot enhances accessibility through natural language interaction.

The system is not intended to replace medical practitioners but aims to support early awareness, guide users, and encourage timely clinical consultation.

6.2 Future Scope

- Integration with real-time clinical test inputs
- Enhanced NLP processing for deeper symptom understanding
- Deployment as a mobile app for wider reach
- Adding more medical datasets for multi-disease screening
- Connecting to cloud-based EHR systems for secure health monitoring

REFERENCES

Taken open source from kaggle-

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